

LUNCHEON HILL, Australia

WMO: 959560

Lat: 41.149S

Lon: 145.152E

Elev: 344

StdP: 97.26

Time Zone: 10.00 (AUS)

Period: 01-19

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
8	1.5	2.3	-0.1	3.9	3.3	0.7	4.1	3.6	14.3	9.3	12.8	9.2	5.2	190	0.735

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
2	9.2	25.6	17.8	23.5	17.0	21.7	16.2	18.7	23.6	17.8	22.0	16.9	20.3	6.0	0

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	
17.1	12.7	19.9	16.2	12.0	19.3	15.4	11.4	18.6	54.6	23.7	51.5	22.0	48.8	20.4	23.6

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max		
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(4) 14.4	12.2	10.8	DB	0.3	30.2	0.7	2.7	-0.2	32.2	-0.6	33.7	-0.9	35.2	-1.4	37.2	(4)
(5)			WB	-0.2	21.1	0.7	1.2	-0.7	22.0	-1.0	22.6	-1.4	23.3	-1.9	24.1	(5)

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec		
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(6) Temperatures, Degree-Days and Degree-Hours	DBAvg	10.9	14.8	15.0	14.0	11.8	9.5	8.0	7.3	7.4	8.3	9.8	11.9	13.1	(6)	
	DBStd	3.84	3.45	3.12	3.29	2.65	2.15	1.82	1.64	2.00	2.21	2.70	3.17	3.22	(7)	
	HDD10.0	398	2	1	3	11	36	62	83	83	61	36	13	7	(8)	
	HDD18.3	2750	122	101	142	197	274	308	341	339	302	264	196	166	(9)	
	CDD10.0	721	150	141	126	64	20	3	1	3	9	31	69	103	(10)	
	CDD18.3	29	11	7	6	0	0	0	0	0	0	0	2	3	(11)	
	CDH23.3	213	96	39	34	1	0	0	0	0	0	2	12	28	(12)	
	CDH26.7	45	26	6	7	0	0	0	0	0	0	0	1	4	(13)	
(14) Wind	WSAvg	6.0	6.2	6.2	5.8	5.7	5.9	5.6	5.9	5.8	6.3	6.2	6.2	6.2	(14)	
(15) Precipitation	PrecAvg	1485	64	59	87	101	151	160	193	200	157	122	96	88	(15)	
	PrecMax	1966	163	155	143	201	288	297	300	424	242	202	149	165	(16)	
	PrecMin	1131	10	10	29	35	50	66	89	83	52	13	19	20	(17)	
	PrecStd	182	37	36	33	42	62	55	49	79	46	46	39	31	(18)	
(19) Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	30.3	26.8	27.4	21.1	16.9	13.5	12.4	14.5	17.2	22.1	25.2	26.9	(19)	
		MCWB	19.9	18.7	18.5	16.0	13.8	11.7	10.5	11.5	13.1	14.7	16.9	17.8	(20)	
	2%	DB	25.8	24.2	23.3	18.7	14.6	12.2	11.3	12.5	14.6	18.5	21.9	23.3	(21)	
		MCWB	17.9	17.4	17.3	14.8	12.6	10.7	10.1	10.4	11.4	13.3	15.7	16.6	(22)	
	5%	DB	23.1	22.3	21.0	16.9	13.4	11.4	10.5	11.2	13.1	16.3	19.5	20.8	(23)	
		MCWB	16.8	16.7	16.2	13.9	12.1	10.2	9.5	9.7	10.5	12.3	14.5	15.4	(24)	
	10%	DB	20.6	20.4	18.9	15.5	12.6	10.7	9.8	10.3	11.6	14.5	17.3	18.8	(25)	
		MCWB	15.7	15.8	15.3	13.4	11.6	9.8	9.0	9.1	9.7	11.4	13.6	14.6	(26)	
(27) Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	20.7	20.0	19.1	16.7	14.3	12.4	11.3	11.8	13.6	15.7	18.0	19.2	(27)	
		MCDB	27.2	25.8	25.0	19.5	15.7	13.1	11.7	13.5	16.2	19.6	23.4	24.1	(28)	
	2%	WB	18.7	18.5	17.9	15.4	13.1	11.2	10.4	10.8	12.1	14.1	16.2	17.4	(29)	
		MCDB	24.1	22.4	22.2	17.6	13.9	11.7	10.9	12.0	14.0	16.9	20.3	21.6	(30)	
	5%	WB	17.5	17.5	16.9	14.6	12.4	10.6	9.8	10.0	11.0	13.0	15.2	16.1	(31)	
		MCDB	21.9	20.8	19.9	16.2	13.2	11.1	10.2	10.9	12.4	15.6	18.7	19.3	(32)	
	10%	WB	16.4	16.6	15.7	13.7	11.7	10.0	9.2	9.3	10.1	11.8	14.0	15.0	(33)	
		MCDB	20.0	19.6	18.5	15.2	12.5	10.5	9.6	10.1	11.3	14.2	17.0	18.5	(34)	
(35) Mean Daily Temperature Range	5% DB	MDBR	9.6	9.2	8.1	6.4	5.0	4.5	4.4	5.3	6.2	8.0	8.5	8.6	(35)	
		MCDBR	13.4	11.8	11.1	8.2	5.8	5.0	5.0	6.2	8.2	10.9	12.2	12.1	(36)	
	5% WB	MCWBR	6.7	5.9	5.7	4.9	4.3	3.9	4.0	4.5	5.3	6.4	6.8	6.5	(37)	
		MCWBR	12.0	10.3	10.0	7.0	5.1	4.1	4.3	5.5	7.2	9.6	11.0	10.5	(38)	
(40) Clear-Sky Solar Irradiance	taub		0.335	0.327	0.319	0.315	0.309	0.298	0.299	0.330	0.340	0.331	0.335	0.327	(40)	
		taud	2.531	2.571	2.594	2.582	2.580	2.602	2.603	2.492	2.461	2.502	2.507	2.551	(41)	
	Ebn at Noon		993	976	941	876	812	790	816	841	900	960	987	1007	(42)	
		Edh at Noon	108	99	88	77	66	59	63	83	99	105	110	107	(43)	
(44) All-Sky Solar Radiation	RadAvg		6.58	5.76	4.26	2.91	1.83	1.46	1.59	2.35	3.45	4.95	6.02	6.51	(44)	
	RadStd		0.43	0.31	0.22	0.20	0.14	0.13	0.11	0.19	0.22	0.31	0.40	0.47	(45)	

Historical Trends

		DBAvg	Heating		Cooling			Degree-Days			
			99% DB	99% DP	1% DB	1% WB	1% DP	HDD10.0	HDD18.3	CDD10.0	CDD18.3
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)
(46) Station Only		N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
(47) Regional (18 neighbors)		+0.27	N/A	N/A	+0.65	+0.41	+0.45	N/A	-80	+90	+11

Nomenclature: See separate page